

Predicting and Understanding Airline Booking Surges using Social Media

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NOTE: Some terms or values have been redacted or generalized at the employer's request.

I. PROJECT SUMMARY

This project aims to leverage various social media data signals to predict future booking surges in flight activity. The main motivation is predicting if demand shocks in the future can improve demand planning and pricing strategy to be more adaptive to upcoming changes. Hence, the airline would have more flexibility adjusting pricing and still reliably receive full occupancy to maximize revenue per flight. We can enrich the flight booking data with online engagement indicators from several sources. Measures such as Reddit virality, YouTube views and interactions, and Wiki page views could collectively serve as useful leading signals for predicting booking surges.

II. BUSINESS JUSTIFICATION

The rise of social media has launched a new phenomenon in the realm of traveling called viral travel. If you have ever seen a YouTube short or TikTok of a beautiful tourist spot in Japan and wanted to book a flight there, you are not alone.

There is a foundation of prior research centered around learning how social media can influence travel behavior. Social media influencers have engendered the rise and fall of many viral trends with their strong followings, which is indicative of the strength social media has on sparking trends that quickly snowball into mass interest and action. In the scope of tourism, captivating social media content has been shown to guide travel intentions [1] [6].

The power of online media to inspire travel is only becoming more prominent, yet the travel industry is still lagging behind on capitalizing on this new viral travel. This research supports the premise that higher engagement and interaction for a destination/location online can be a precursor for booking demand. Travel content and news can quickly increase awareness of a destination in a short amount of time, so if these social media signals were identified, it could allow airlines to plan and be more responsive to a demand surge.

The goal of this project is to develop a performant model that can predict surges in flight bookings using social media posts and other online popularity signals. The main social media model inputs we had available to us were YouTube and

Reddit, which provide a look into what the hottest destinations are and what is getting the most views and engagement. We also supplemented those social platforms with Wikipedia page views and relative Google Search trend relevance to provide broader context on destination virality and public attention. In particular, daily Wikipedia page views per city serve as a proxy for destination popularity, and Google Search trends offer a normalized measure of weekly search interest for a location. Together, these features help capture whether a destination is gaining online presence before an increase in bookings is observed.

III. LITERATURE REVIEW

Researchers have previously attempted to solve similar problems with similar goals. This research provides important learnings that inform our approach and method. Y. Yao, Y. Li, Y. Liu, and Y. Liu have successfully applied social media sentiment to predict stock market trends [2], providing confidence in the predictive power of social media. Here social media acted as a leading signal in an analogous system [2].

We also learn that sentiment analysis provides superior predictive information for future price movements (over raw volume / engagement) [3]. Reviewing similar literature, we see that consistent sentiment, over multiple days, leads to more accurate predictions [1].

Iftikhar and Khan had success extracting topics and combining sentiment and word analysis from multiple sources for demand forecasting based on social media big data analytics. They found that their emotion enhanced model improved forecasting accuracy over all tests [4].

IV. DATA

A. YouTube

The YouTube Data API was used to collect information on 100k videos and shorts, including their complete metadata as well as related channel data. YouTube granted an education request for additional API credits, which allowed video data to be collected over the course of a week. Data is stored in a human-readable CSV format.

There are approximately 60k YouTube Videos included and 40k YouTube Shorts, with different engagement dynamics. Videos were gathered using a list of about 300 search terms,

fetching data covering the flights dataset period (as well as 6 months prior to allow for leading effects).

B. Reddit

Obtaining Reddit data presented several challenges that resulted in a few methods being tried before a suitable solution was found. The initial idea was to access Reddit data through the official Reddit API. However, the access request was rejected despite using institutional credentials, and the appeal was denied. A web scraping approach was subsequently attempted as an alternative. However, Reddit enforces strict rate limiting and bot detection on its public facing pages, and the JSON endpoint available without authentication restricts access to the 1,000 most recent posts per subreddit. For subreddits with high posting frequency this window covers only a few days of history, making it insufficient for time series analysis spanning over a year. The final and successful approach was to source Reddit data through Academic Torrents, a platform that distributes large research datasets including Reddit post dumps. Complete post histories for five travel-focused subreddits were downloaded covering all posts from January 2024 through December 2025. The following are the subreddits that were used:

- r/EuropeTravel – European destination discussions, trip reports, and travel advice
- r/UKTravel – United Kingdom focused travel content
- r/SoloTravel – Solo traveller experiences and destination recommendations
- r/Travel – General travel community, one of the largest travel subreddits on Reddit
- r/Backpacking – Budget and independent travel discussions

C. Wikipedia and Google Trends

Wikipedia and Google Trends data were used to gauge changing public interest in destinations over time. Destination airport and rail codes were mapped to city-level Wikipedia titles, supported by manual overrides for edge cases. Daily Wikipedia page views were retrieved through the Wikimedia REST API as a direct measure of destination attention. Then, weekly Google Trends data was collected using pytrends for destination-specific search phrases such as “city flight” for web search and “city travel” for YouTube search. Anchor terms were included in each query, so the relative popularity would be comparable. Together, this data creates online interest signals for downstream analysis to compare against bookings.

D. Flight Bookings

Over 1 million European flight booking transactions were provided by Lufthansa covering the period of 1 year. This forms the basis of our dataset, aggregated to the daily level per IATA destination airport before being merged with the previous datasets.

Data covers December 2024 to December 2025 and is highly seasonal peaking in European summer, the limited

scope of the data makes monthly STL decomposition challenging. For our model, we explicitly consider bookings by outgoing passengers to specific destination airports, noting that this is the primary driver for travel. For the model, we also filter out airports with less than 30k annual passengers and less than 25% daily YouTube coverage to reduce noise related to low information destinations.

E. Exploratory Data Analysis

Lead time analysis showed that the median booking lead time (the number of days between transaction date and departure date) was relatively shorter than the mean reflecting a right-skewed distribution. The majority of bookings occurred within a typical planning window prior to departure, with a long tail extending much further out for advance planners. A significant concentration of bookings occurred close to the departure, suggesting both last minute travelers and planned journeys coexist in the dataset. This distribution was important context for lag analysis: a Reddit signal appearing 7-28 days before a booking surge falls within the active booking window for most travelers, confirming that social media signals have a commercially realistic lead time. STL decomposition was applied to the aggregate daily booking series to decompose it into three components: seasonal, trend, and residual. Using a period of 7 days to capture the weekly booking cycle, the decomposition revealed:

- A strong weekly seasonal component with swings of approximately 40,000 passengers per day driven by day-of-week booking preferences
- A broad trend component peaking at approximately 195,000 passengers per day in early 2025, declining gradually through the year
- A residual component with occasional spikes of 25,000 passengers representing genuine unexplained demand variation

The seasonal component was notably large relative to the residual meaning weekly cycles alone account for fluctuations larger than the genuine demand shocks the model aims to predict. This finding directly motivated the deseasonalization approach applied later in the analysis.

Social media coverage is good across the destination airport bookings dataset, providing a strong foundation for the model and further analysis.

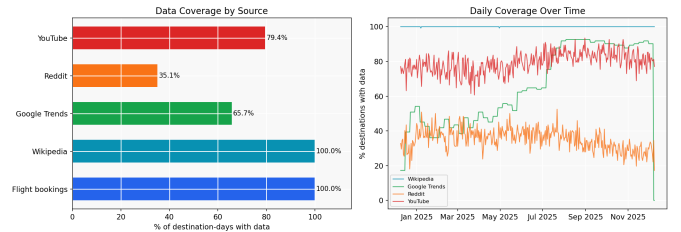


Fig. 1. Data coverage by source

We’re able to note that the coverage provided by the Reddit data is less than YouTube, and this will later have implications

in modelling, this is primarily due to the constraints faced when acquiring the Reddit data and the limited set of subreddits used. This could be expanded in future.

F. Data Processing and NLP Pipeline

1) YouTube:

- NLP Location Extraction - We perform multi-layer matching to extract relevant locations.

Layer 1: spaCy NER (en_core_web_trf) for named entity extraction → IATA lookup.

Layer 2: sentence-transformer (all-MiniLM-L6-v2) semantic embedding matching against rich airport descriptions — catches landmarks, events, concerts, food references without hard coded keywords. This allows a video about Oktoberfest to be matched to Munich, or a video about Sziget to be matched with Budapest.

Layer 3: RapidFuzz fuzzy tag matching for typos and variations.

- Sentiment Analysis - We utilise cardiffnlp/twitter-roberta-vase-sentiment-latest transformer model which produces continuous (-1, 1) sentiment scores for videos based on the text in the title, description and tags. This is useful for topic level sentiment decomposition.
- Demand Intent Classification - Keyword based classification of content into booking intent based on curated phrase lists. Applied consistently to YouTube and Reddit data.
- Travel Theme Extraction - 30 theme taxonomy defined by over 1,000 keywords. Applied consistently to YouTube and Reddit data.
- Commercial Intent - Monetisations, booking sites, and commercial intent is flagged based on keyword matching.

2) *Reddit Enrichment:* Each Reddit post was processed using a pre-trained sentiment analysis model to produce a compound sentiment score ranging from -1 (most negative) to +1 (most positive). To accurately identify locations mentioned in social media text, we enriched destination metadata with alias mappings for relevant airport–city pairs and used spaCy for Named Entity Recognition to detect cities. Sentiment was computed using VADER, which returned both a sentiment label and compound score for each post, along with detected destinations including their name, country, and ISO code. The sentiment compound score captures the overall emotional direction of a post, and posts were labelled as positive, negative, or neutral based on standard compound score thresholds. The raw Reddit data was then cleaned and processed to produce three engagement and sentiment metrics, aggregated at the daily level per destination. Engagement captures the volume and reach of Reddit activity as a weighted sum of upvotes, comments, and cross-posts, where comments and cross-posts receive higher weights because they represent more deliberate forms of engagement:

$$\text{Engagement} = \text{Score} + 2 \cdot \text{Num_comments} + 3 \cdot \text{Cross_posts}$$

Engagement score applies a log transformation to compress the extreme values produced by viral posts. Weighted sentiment

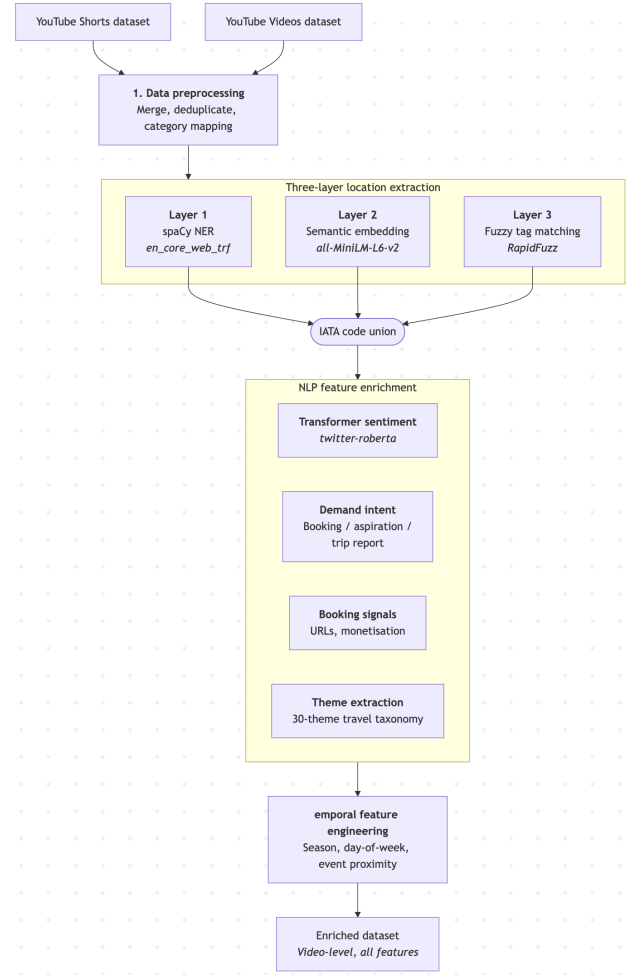


Fig. 2. The data processing pipeline used to extract text features and information from the YouTube dataset.

combines the engagement score with the sentiment compound score of each post, capturing both the intensity and emotional direction of Reddit activity.

$$\text{Weighted Sentiment} = \text{Engagement} \times \text{Sentiment Compound}$$

After destination extraction from text or title of the Reddit posts, the destination names were contained as free text strings while the LHG booking data used ISO alpha-2 country codes. Joining the two datasets required a robust matching pipeline to handle naming variations, alternative country names, and pycountry vocabulary differences. A four-stage matching pipeline was implemented, applied in priority order:

- **Stage 1 – Manual overrides:** A curated dictionary of known mismatches was applied first. This captured cases where automated methods were known to fail, such as “Turkey”, which pycountry resolves to “Türkiye” following the country’s official name change in 2022.
- **Stage 2 – Exact normalised matching:** Both sides were lowercased and stripped of whitespace before compari-

son. This resolved straightforward cases like “Austria” matching “AT” through `pycountry`’s full name lookup.

- **Stage 3 – ISO alpha-3 bridge:** Reddit destination strings were resolved to ISO alpha-3 codes via `pycountry`’s fuzzy search, then bridged to the alpha-2 codes in the LHG data. This handled alternative country names such as “Deutschland”, “Republic of Albania”, and formal UN names.
- **Stage 4 – Fuzzy string matching:** `RapidFuzz`’s `WRatio` scorer was applied against the full country name list with a threshold of 95. This threshold was deliberately set high after initial testing at 85 produced numerous false positives, most notably “Australia” matching to “Austria” with a score of 88. All fuzzy matches were audited manually before acceptance.

The pipeline successfully matched 45 out of 221 unique Reddit destinations to LHG country codes (41 exact, 3 via `pycountry`, 1 manual). The 176 unmatched destinations were all correctly non-European countries outside LHG’s dataset and were excluded from subsequent analysis. This matched 41.9% of Reddit rows to an LHG country, with the remaining 58.1% representing travel discussion about destinations outside this project’s scope.

V. ATTEMPT 1 - REDDIT SIGNAL & FLIGHT BOOKING INTERACTIONS AND RELEVANCE SCORING

Both the booking and Reddit panels were sparse after aggregation since not every country had entries for every date. Rolling window calculations applied to sparse series produce incorrect results because windows are computed over fewer days than intended, silently understating rolling sums and means. To prevent this, a complete date spine was constructed for each country covering the full date range, and missing dates were filled with zero values. This step expanded the Reddit panel from 22,706 to 30,012 rows, with the additional rows representing country-days where no Reddit posts were recorded.

A. Feature Engineering

Rolling window aggregations for the LHG dataset were computed at 7, 14, and 28-day windows for daily passenger counts per country, producing both rolling sums and rolling means. These capture demand at different time horizons: the 7-day window reflects recent weekly demand, while the 28-day window captures the monthly baseline. Similarly for the Reddit datasets, rolling window aggregations at 7, 14, and 28-day windows were applied to total engagement score and weighted sentiment, producing rolling sums and rolling means that smooth day-to-day noise and capture sustained activity patterns.

Sentiment momentum was defined as the difference between the 7-day and 28-day rolling mean sentiment. A positive value indicates that recent sentiment is improving relative to the longer baseline, capturing destinations that are trending positively rather than those that are persistently positive.

Engagement momentum was defined analogously for engagement scores, capturing whether Reddit activity for a destination is growing or declining relative to its own recent baseline.

Engagement z-score measured whether today’s Reddit engagement is unusually high relative to the country’s own rolling 28-day baseline, identifying viral or spike moments in social media activity.

Consistent sentiment score was included for a comparison between raw sentiment and sustained sentiment. The feature combines two dimensions: the proportion of the last 28 days with positive sentiment, multiplied by the 28-day rolling mean sentiment level:

A high score requires both sustained positivity (many positive days) and meaningful sentiment level. Based on this method, a destination with one spike of positive sentiment would score much lower than one with consistent positive discussion over the full month.

Lag features were created by shifting each Reddit feature forward in time by 7, 14, 21, and 28 days. Shifting a Reddit signal forward by 14 days places the value from 14 days ago alongside today’s bookings, allowing the model to directly ask whether historical Reddit activity predicts current booking behaviour. This produced 68 lag features (17 base features $\times$ 4 lag windows) covering the full range of plausible customer response times identified in the literature.

After adding these additional features, the booking and Reddit datasets were joined on country and date using an inner join, retaining only dates where both booking and Reddit data were available. The resulting panel contained 15,006 rows across 41 countries covering December 9, 2024, to December 9, 2025, with 109 total columns.

B. Correlation Analysis

The Cross-Correlation Function (CCF) was computed between each Reddit signal and daily bookings across lags from 0 to 42 days. Unlike computing correlation at a single fixed lag, the CCF evaluates the relationship at every lag simultaneously, allowing the data to determine where the predictive relationship is strongest. The analysis was conducted per country and then aggregated across all markets to identify consistent patterns.

Spearman rank correlation was used in preference to Pearson correlation because the relationship between social media signals and booking volumes is unlikely to be linear – viral posts can produce extreme engagement values that would distort a linear correlation measure. Spearman’s rank-based approach is robust to such outliers.

From the EDA we saw that both Reddit engagement and bookings peak in summer. This shared seasonal cycle produces a high correlation at every lag because both series are elevated at the same time of year regardless of any genuine temporal leading relationship. To isolate genuine predictive signal, STL deseasonalization was applied to the daily booking series for each country individually, removing the weekly seasonal

component. Spearman correlations were then recomputed between the deseasonalized booking series and the lagged Reddit features.

A complementary event-based analysis examined whether Reddit engagement spikes predict booking spikes. For each instance of a country’s Reddit engagement exceeding 2 standard deviations above its rolling baseline, the probability of a booking spike occurring within subsequent windows of 7, 14, 21, and 28 days was computed.

A CCF comparison was conducted between sentiment momentum and consistent sentiment score to evaluate the two sentiment approaches. Both signals were tested against daily bookings across lags from 0 to 42 days, with the mean correlation across all countries plotted alongside confidence intervals to assess both the strength and consistency of each signal.

C. Relevance Score

The relevance score is a composite daily signal computed per country that estimates the likelihood of above-average booking demand in the following 7-28 days, based on current Reddit engagement and sentiment patterns. It is designed to answer the question: is this destination currently receiving unusually elevated social media attention, in a pattern historically associated with booking surges?

All input features were normalized within each country’s own historical range (0-1 MinMax scaling) before scoring. This per-country normalisation ensures that the score measures deviation from each market’s own baseline - an elevated score for Kosovo means Kosovo’s Reddit buzz is in the top portion of its own historical range, not that it has more absolute engagement than France.

Three scoring approaches were developed with deliberately different weighting philosophies to enable comparison and understand which aspect of the Reddit signal drives predictive power.

Approach A - Weighted Linear Score

Weights were assigned directly proportional to the deseasonalized Spearman correlations measured in the correlation analysis, making this the most evidence-grounded approach. This approach prioritizes recent engagement volume (7-day rolling) and sustained engagement (28-day rolling) roughly equally, with consistent sentiment as a meaningful secondary factor.

Approach B - Sentiment-First Score

This approach tests whether the quality and direction of sentiment matters more than raw engagement volume. Consistent sentiment received 40% weight, sentiment momentum 20%, with engagement features reduced to 40% combined. If Approach B outperforms A, it would indicate that why people are discussing a destination is more predictive than how much they are discussing it.

Approach C - Engagement Momentum Score

This approach focuses entirely on change rather than level, placing 50% weight on positive engagement momentum. This would test whether Reddit buzz is currently growing above its



Fig. 3. STL Decomposition of Aggregated Passenger Sum Daily Booking

own baseline. Only upward momentum was scored (negative momentum received zero weight). This is the most forward-looking approach, designed to identify destinations early in a buzz cycle before they reach peak engagement.

D. Results and Evaluation

Figure 3 above presents the decomposition of the aggregate LHG booking series into its seasonal, trend, and residual components. The seasonal panel confirms a strong and consistent weekly cycle throughout the year, with swings of approximately 40,000 passengers per day. This cycle is driven by day-of-week booking preferences and is present across all 52 weeks of the dataset without meaningful variation in amplitude. The trend panel shows aggregate daily demand peaking at approximately 195,000 passengers in January-February 2025 (the primary planning window for summer travel) before declining through the year to approximately 120,000 passengers by December 2025. The residual panel is largely contained within 25,000 passengers, with a small number of isolated spikes which likely correspond to specific demand events. Crucially, the seasonal component is larger in magnitude than the residual, meaning day-of-week effects alone produce fluctuations greater than the genuine unexplained demand variation the model aims to predict. This observation established the necessity of deseasonalization before any correlation analysis.

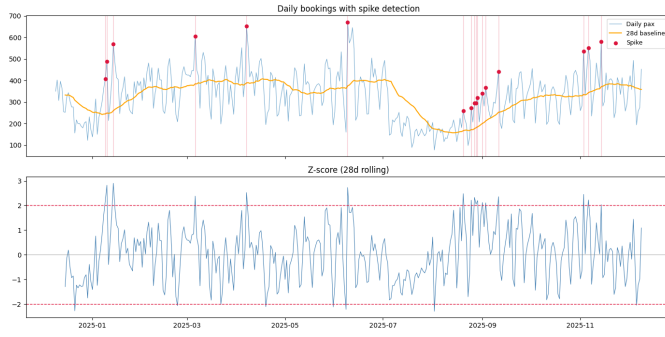


Fig. 4. Aggregated Passenger Sum Daily Bookings with Spike Detection for Sample Country K

Per-country spike detection identified 263 positive booking spike days across 41 countries over the 12-month period, averaging approximately 6 spike days per country per year. Country K recorded the most spikes at 17, which can be seen above in Figure 4, reflecting the higher relative volatility of a small market where even moderate absolute increases exceed the 28-day rolling standard deviation. The aggregate spike rate of 1.7% of country-days established the baseline against which the relevance score's predictive lift would later be measured.

Figure 5 shows the CCF (Cross Correlation Function) curves for seven Reddit signals aggregated across all 41 countries. The raw daily engagement, weighted sentiment, and engagement z-score signals exhibit pronounced oscillating patterns with peaks repeating at 7-day intervals. This is a weekly seasonality artifact meaning both Reddit activity and bookings have strong weekly cycles driven by day-of-week behaviour, and the CCF picks up this shared rhythm rather than a genuine multi-week predictive relationship. These oscillating curves should not be interpreted as evidence of periodically recurring predictive windows; the meaningful information lies in the envelope of the waves rather than the individual peaks.

The rolling aggregated signals (engroll7 and engroll28) produce smooth, monotonically decreasing CCF curves that are free of the weekly oscillation, confirming that the artifact was driven by daily signal granularity rather than the underlying relationship. The engroll28 signal achieves the highest aggregate peak correlation at $r=0.184$ with optimal lag at 0 days, while engroll7 peaks at $r=0.149$ at lag 2 days. The apparent lag-0 optimum for the rolling signals is a product of the shared summer seasonality discussed below and is corrected through deseasonalization.

Sentiment momentum produced near flat correlations across all lags (peak $r=0.025$ at lag 36 days), indicating that changes in sentiment direction carry minimal predictive information in the aggregate. Consistent sentiment produced a smoothly decaying curve starting at $r=0.132$ at lag 0, confirming it is a more stable signal than momentum but one also requiring deseasonalization to isolate genuine predictive content.

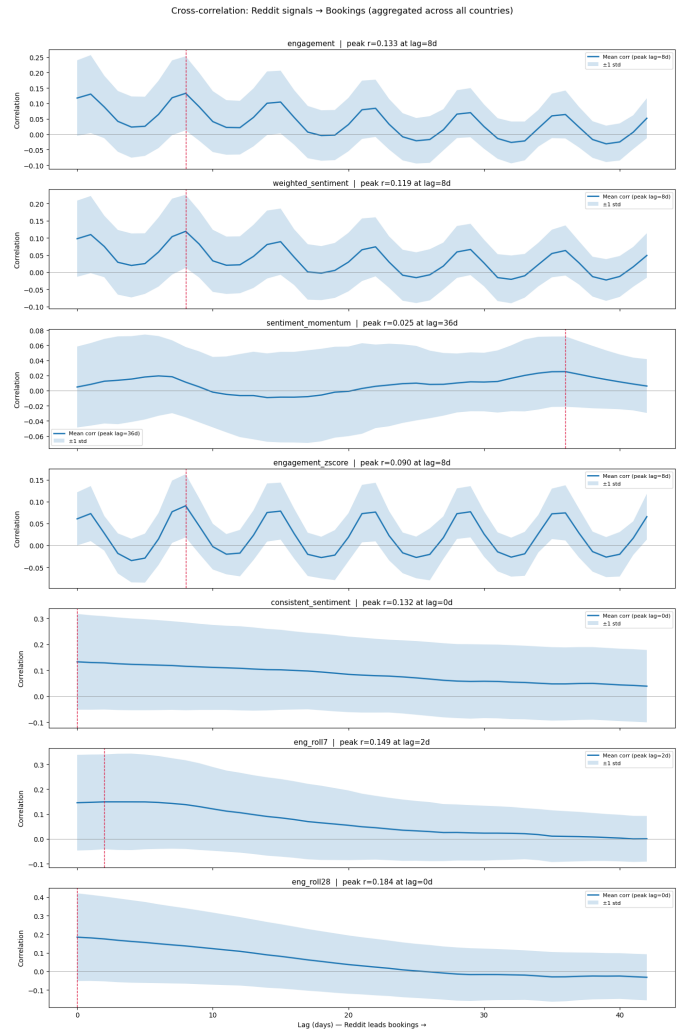


Fig. 5. Cross Correlation of Reddit Signals with Bookings

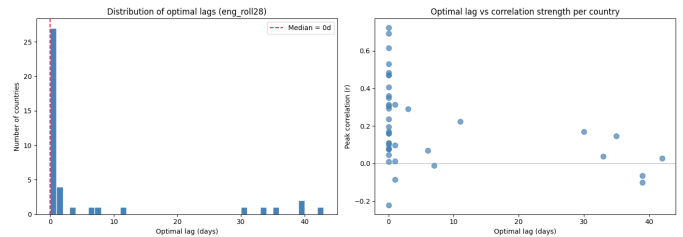


Fig. 6. Optimal Lag Distribution

Figure 6 shows the per-country lag analysis for the engroll28 signal. The histogram seen in Figure 6 shows that 27 out of 41 countries have an optimal lag of 0 days and the median optimal lag is 0. This concentration at lag 0 is the seasonal confound in action meaning when both Reddit engagement and bookings peak in summer, the CCF algorithm identifies same-day correlation as strongest because both series are elevated simultaneously due to the shared underlying cycle.

The scatter plot on the right of Figure 6 reveals the more nuanced picture. Countries clustered in the upper left quadrant

have both high peak correlations and near-zero optimal lags, driven largely by shared seasonality. More informative are the outlier countries at non-zero lags: one country has an optimal lag of 11 days with $r=0.22$, one peaks at lag 30 with $r=0.17$, and another peaks at lag 35 with $r=0.15$. These represent genuine cases where Reddit engagement measurably precedes bookings by multiple weeks, independent of seasonal co-movement.

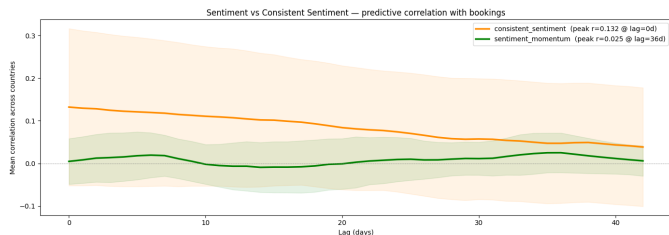


Fig. 7. Cross Correlation of Sentiment Momentum and Consistent Sentiment

Figure 7 shows the CCF comparison between sentiment momentum and consistent sentiment score. We can see that consistent sentiment substantially outperforms sentiment momentum at every lag from 0 to 42 days. Consistent sentiment maintained a mean correlation of approximately $r=0.13$ across all lags while sentiment momentum remained essentially flat near zero throughout, with a nominal peak of $r=0.025$ at lag 36 days. The wide confidence band around consistent sentiment reflects meaningful country-level variation, but the mean effect is robust and directionally consistent across markets. The result confirms that how long a destination has been discussed positively is a far more informative signal than whether sentiment is currently improving which is a finding with direct implications for the design of the relevance score.

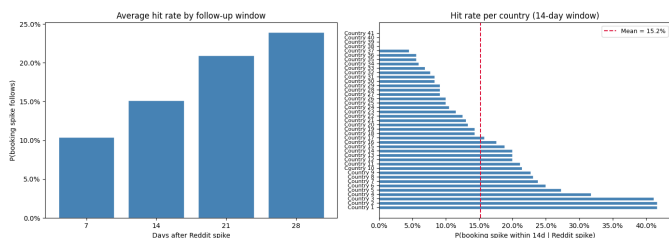


Fig. 8. Hit Rate by Lag and Country

The base rate of a positive booking spike on any given country-day was 1.7%. Following a Reddit engagement spike for the same country, the probability of a booking spike occurring within the following 7, 14, 21, and 28 days was 10.5%, 15.2%, 20.9%, and 23.9% respectively as seen in Figure 8 above. At the 14-day window this represents approximately a 9-fold increase in booking spike probability compared to the base rate. The monotonic increase across windows confirms that the Reddit signal requires time to translate into booking behaviour, consistent with the lead time findings from the EDA. Country-level analysis revealed substantial heterogeneity. Three countries showed hit rates above 40% at the 14-day

window, meaning Reddit engagement spikes in these markets are followed by booking spikes in more than 4 out of 10 cases within two weeks. At the other extreme, two countries showed near-zero hit rates, indicating that Reddit discussion about these destinations does not reliably precede booking surges. This may be because the Reddit audience discussing these destinations is not aligned with LHG's customer base or due to the data collection limitation process, as the selected subreddits may not fully capture the relevant online discourse for these locations.

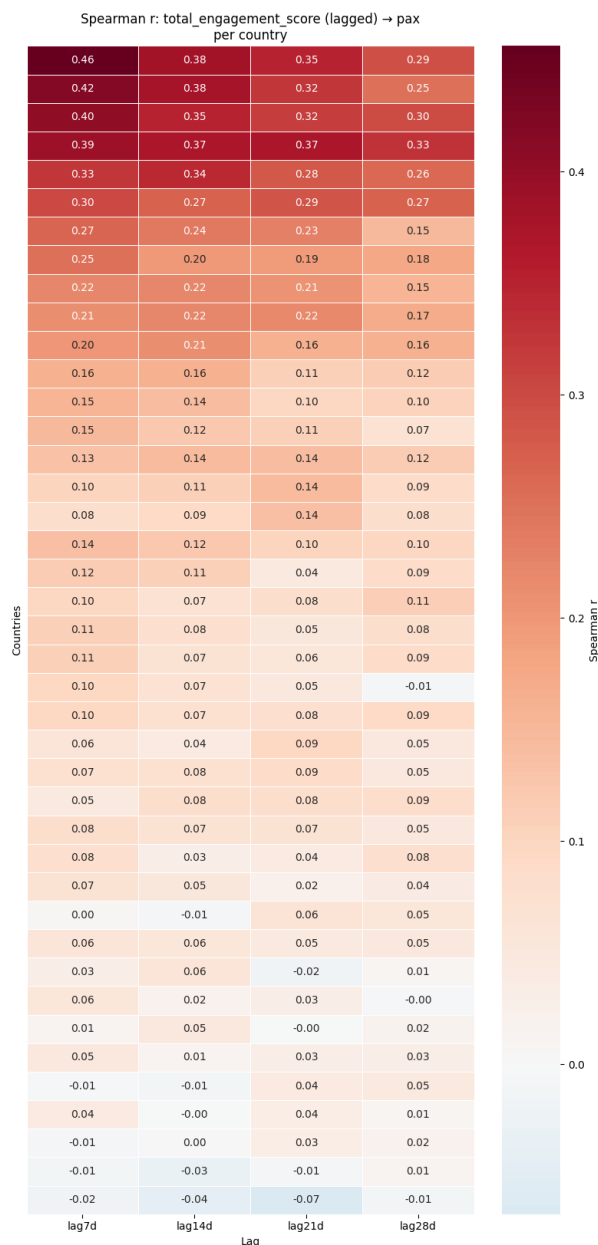


Fig. 9. Spearman Correlation Across Different Lags

Figure 9 shows the per-country Spearman correlations between total engagement score at lags 7, 14, 21, and 28 days and daily bookings. Country A ($r=0.46$ at lag 7) and Country

B ($r=0.42$), Country C ($r=0.40$), Country D ($r=0.39$), and Country E ($r=0.33$) are the five strongest markets. All are Mediterranean or Southern European leisure destinations characterised by high Reddit discussion volume and aspirational travel appeal. The correlation decays with lag for most of these countries for example, Country A declines from 0.46 at lag 7 to 0.29 at lag 28, and Country B from 0.42 to 0.25, providing early evidence of a genuine temporal leading relationship even before deseasonalization is applied. At the lower end of the table, Country F, Country I, Country J, and Country K show near-zero or negative correlations across all lags. These markets have insufficient Reddit discussion volume to generate a reliable signal, a finding that shaped the decision to evaluate the relevance score separately for high-signal and all-country subsets.

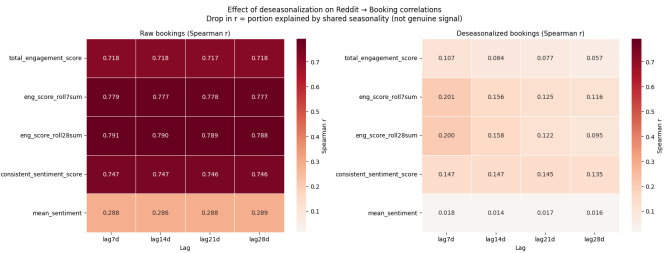


Fig. 10. Comparison of Spearman Correlation between Raw and Deseasonalized Bookings

Figure 10 presents the most critical methodological result of the correlation analysis. The left heatmap shows raw Spearman correlations between lagged Reddit features and daily bookings, with values ranging from 0.71 to 0.79 for engagement features. These values are nearly identical across all four lags (7d, 14d, 21d, 28d). This is a pattern inconsistent with a genuine temporal relationship, where correlation should decay as the lag increases if Reddit truly leads bookings. The right heatmap shows the same correlations after applying per-country STL deseasonalization to the booking series. The values drop substantially across all features as can be seen in Table I below.

TABLE I
CORRELATION COMPARISON BEFORE AND AFTER DESEASONALIZATION (LAG7)

Feature	Raw r (I7)	Deseasonalized r (I7)	Drop
engscoreroll7sum	0.779	0.201	0.578
engscoreroll28sum	0.791	0.200	0.591
consistentsentimentscore	0.747	0.147	0.600
totalengagementscore	0.718	0.107	0.611
meansentiment	0.288	0.018	0.270

Approximately 60% of the raw correlation was attributable to shared seasonality rather than genuine predictive signal. After deseasonalization, the decay pattern with lag is clearly restored. Engscoreroll7sum declines from $r=0.201$ at lag 7 to $r=0.116$ at lag 28, confirming the presence of a genuine temporal leading relationship that the raw correlations had obscured. Mean sentiment collapses to $r=0.018$, which is

statistically negligible, confirming that sentiment level carries almost no predictive information once the seasonal cycle is removed. The deseasonalized values were used as the empirical basis for all subsequent relevance score weighting. The three relevance score approaches were first evaluated against the strictest possible target: predicting rare positive booking spikes (z -score > 2.0) within 14 days. With a base rate of only 1.7% spike days this was an extremely hard classification task; a random predictor flagging all country-days as positive would achieve precision of 11.1%.

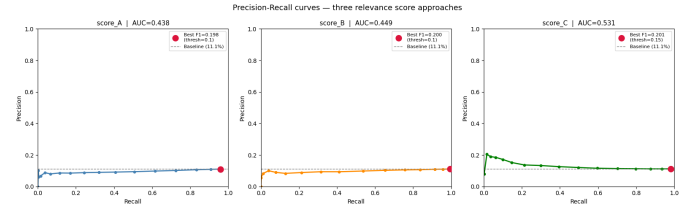


Fig. 11. Precision Recall Curves

Figure 11 shows the PR curves for all three approaches. Approaches A and B produce curves that hug the baseline throughout, with neither able to lift precision above the base rate at any recall level. Score C achieves a small but meaningful lift at low recall, reaching approximately 20% precision at high thresholds compared to the 11.1% baseline which was roughly doubling the hit rate for its highest-confidence signals.

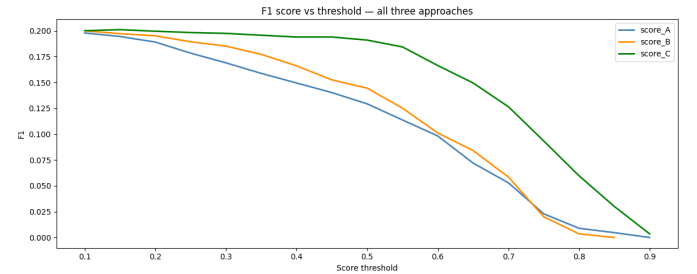


Fig. 12. F1 Scores

Figure 12 shows that Score C maintains meaningful F1 values up to threshold 0.5, while Approaches A and B degrade significantly faster. The best threshold for all three approaches was low (0.10-0.15), indicating the scores do not discriminate sharply enough to support high-confidence flagging on the strict spike target. The AUC results were 0.531 for Approach C, 0.449 for Approach B, and 0.438 for Approach A. Only Approach C beat random on this strict evaluation. This initial result was not treated as a failure but as evidence that the evaluation framework was mismatched to the score's design intent. The score was built to identify destinations with sustained elevated demand momentum, not to predict rare isolated spikes. A single booking spike day could be driven by many factors such as airline schedule changes, public holidays or competitor pricing. All of these have no relationship to Reddit engagement. Evaluating against this target was therefore

deemed overly conservative. The evaluation was repeated with a more appropriate target: predicting above-average booking days (top 33% of daily passenger counts for each country) within a 28-day window. This target has a base rate of 33% and better reflects the score's actual use case which was identifying periods of elevated sustained demand rather than isolated rare events.

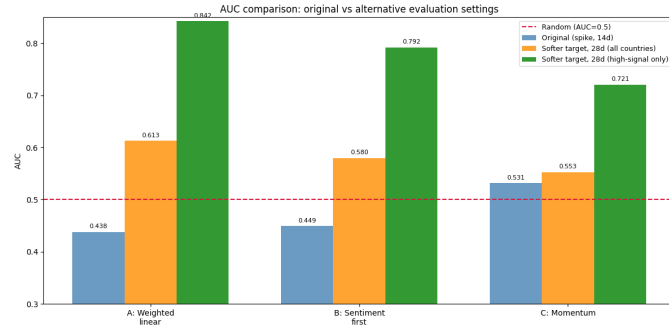


Fig. 13. AUC Comparison

Figure 13 presents the full comparison. Under this alternative framework the results changed substantially. Approach A improved from AUC=0.438 to AUC=0.613 across all countries, and to AUC=0.842 on the eight high-signal countries. This reversal from worst performer on the strict target to best performer on the realistic target, is explained by the nature of the signals. Approach A's sustained engagement features (7-day and 28-day rolling engagement) are well-suited to predicting periods of consistently elevated booking demand, but they do not react sharply enough to catch the sudden isolated spikes that Approach C's momentum focus is designed to detect. Approach B achieved AUC=0.792 on high-signal countries, sitting between A and C. The result for Approach B was in the middle; never the best, never the worst across either evaluation and this confirms that sentiment adds predictive value but should not dominate over engagement volume. Approach C achieved AUC=0.721 on high-signal countries, its relative decline from the strict evaluation reflecting the fact that momentum signals are better calibrated to detect sudden changes than sustained above-average periods. The gap between all-country (A: 0.613) and high-signal-only (A: 0.842) performance quantifies where the Reddit signal is operationally useful. The 0.229-point improvement when restricting to high-signal markets confirms that the score should be deployed selectively rather than uniformly across all 41 European markets.

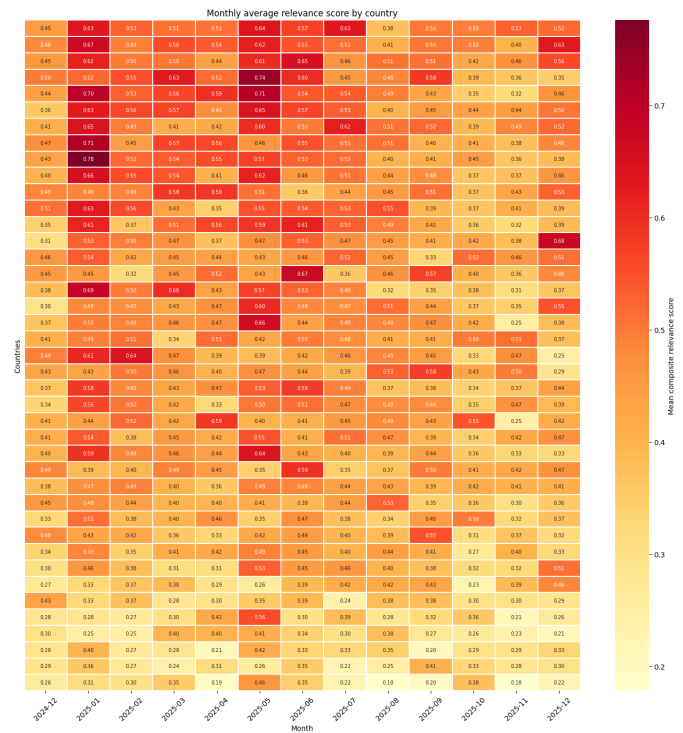


Fig. 14. Monthly Average Relevance Score

Figure 14 presents the composite relevance score averaged by country and month. Several patterns are immediately apparent. January 2025 and June 2025 are universally high-scoring months across almost all markets, possibly corresponding to the New Year travel planning surge and the summer booking season respectively. One country scored 0.74 in June 2025, which was the single highest monthly score in the dataset. This indicates a particularly pronounced above-baseline Reddit engagement surge for that destination during the peak summer planning period. Another country scored 0.68 in December 2025, suggesting elevated social media interest in the run-up to the year-end holiday period. Countries at the bottom of the heatmap, consistently score below 0.30 across all months, confirming their classification as low-signal markets where Reddit discussion volume is insufficient to generate actionable signals.

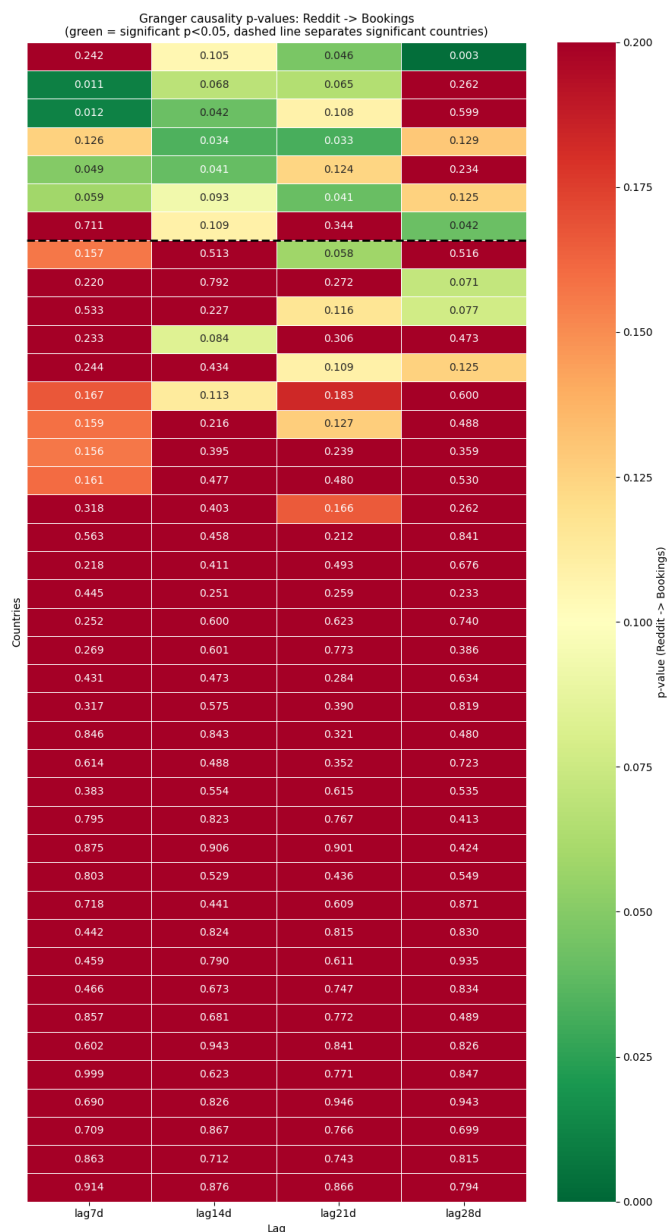


Fig. 15. Granger Causality Across Lags

Figure 15 presents the p-values from the Granger causality tests across all countries and four lag windows. The colour scale uses green to indicate low p-values (statistically significant, $p < 0.05$) and red to indicate high p-values (not significant). The dashed black line separates the 7 countries with at least one significant lag from the 34 countries with no significant causality at any tested lag. Seven out of 41 countries showed statistically significant Granger causality for the Reddit signal to bookings. The lag structure of confirmed causality varied meaningfully across markets. Country M and N showed significance at lag 7 days (the fastest customer response time observed) suggesting Reddit buzz about these destinations translates to bookings within a single week. Country M was significant at lags 14 and 21 days, indicating

a 2-3 week response window. Country N was significant only at lags 21 and 28 days (the longest response cycle in the dataset) suggesting Country N travellers take 3-4 weeks to convert Reddit inspiration into a booking, or that France as a destination is more deliberately planned than impulse purchases. Directionality analysis revealed that five countries showed clean unidirectional causality with Reddit leading bookings and no significant reverse relationship. Two other countries showed bidirectional causality, meaning bookings also significantly predict Reddit engagement for these markets. This bidirectionality likely reflects a feedback loop where travellers post about their trips after booking or after travelling, making the causal interpretation more complex for these markets. It is important to note that the absence of confirmed Granger causality in 34 countries does not imply Reddit has no relationship with bookings in those markets. The Granger test is conservative and requires a strong signal to reject the null hypothesis. For markets with low Reddit discussion volume the test is likely underpowered rather than genuinely negative.

Why Deseasonalization Was Critical

The most important methodological finding of this project was the discovery that approximately 60% of the apparent raw correlation between Reddit engagement and LHG bookings was attributable to shared seasonality rather than genuine predictive signal. Raw Spearman correlations of 0.71-0.79 appeared impressive but were nearly identical across all lag windows. This is a pattern that is statistically impossible if the relationship were genuinely temporal. A real leading indicator produces correlations that decay as the lag increases; a shared seasonal cycle produces correlations that remain flat regardless of lag because both series are simply elevated at the same time of year.

This finding has implications beyond this project. It suggests that social media-to-bookings analyses that do not apply deseasonalization are likely to substantially overstate the strength of their signals. The AUC=0.842 relevance score result reported here is a conservative estimate grounded in deseasonalized data which is the true predictive value above seasonal noise, not an inflated figure that mixes signal with seasonal co-movement. Hence, for similar use cases replication or extension of this analysis should treat deseasonalization not as an optional step but as a prerequisite for credible results.

What the Reddit Signal Actually Captures

The deseasonalized correlation analysis revealed a clear hierarchy of Reddit signal quality. The 7-day rolling engagement sum ($r=0.201$ at lag 7) and 28-day rolling engagement sum ($r=0.200$ at lag 7) were the strongest genuine predictors, both decaying meaningfully with lag, confirming a real temporal leading relationship. Consistent sentiment ($r=0.147$) was a stable background signal. Mean daily sentiment collapsed to $r=0.018$ which essentially has zero predictive value.

This hierarchy tells a coherent story about what Reddit

engagement actually measures in the context of travel. When a destination begins to accumulate sustained Reddit engagement such as multiple posts, discussions, comments over days and weeks, this reflects a growing awareness and interest in that destination within the travel community. This sustained interest translates to bookings over the following 1-2 weeks as users move from inspiration to planning to purchase. A single high-sentiment post, by contrast, represents a momentary spike in visibility that does not necessarily persist long enough to influence booking behaviour at scale.

This interpretation is reinforced by the consistent sentiment finding. The feature that captured whether a destination had been positively discussed for most of the past 28 days outperformed the feature measuring whether sentiment was currently improving. This suggests the booking decision is driven more by an accumulating background of positive awareness. That is a destination gradually building a reputation in travel communities rather than by any single viral moment. This means monitoring the rolling 7-14 day engagement trend is more operationally valuable than reacting to individual post spikes.

Sentiment vs Consistent Sentiment

The comparison between raw sentiment and consistent sentiment was one of the most practically useful findings of the project. The result was unambiguous: mean daily sentiment was essentially useless as a predictor ($r=0.018$ after deseasonalization) while consistent sentiment retained meaningful predictive power ($r=0.147$) that remained stable across all lag windows from 7 to 28 days.

This finding aligns with what is known about consumer decision-making in travel. Booking a flight is not an impulsive decision triggered by a single positive post. A destination needs to capture sustained positive attention in a traveller's social environment over time before it rises to the top of their consideration set. For example, one viral post about Albania may produce a momentary curiosity; twenty consecutive days of positive Albanian travel content creates the sustained awareness that translates into bookings.

The stability of consistent sentiment across lags (0.147, 0.147, 0.145, 0.135 at lags 7, 14, 21, 28) is particularly informative. Unlike engagement features which decay sharply with lag, consistent sentiment remains predictive even at 28 days. This suggests consistent sentiment operates as a slower-moving background signal rather than a short-term trigger. This suggests a destination scoring high on consistent sentiment and showing elevated recent engagement is a more reliable commercial signal than one with elevated engagement alone.

The sentiment-first scoring approach (Approach B), which gave dominant weight to consistent sentiment, achieved $AUC=0.792$ on high-signal countries under the realistic evaluation framework. While this is below Approach A's 0.842, it is a strong result that confirms sentiment information adds value beyond engagement volume alone. The optimal approach is a

combination, which is exactly what Approach A implements through its evidence-based weighting.

The Granger Causality Results

The confirmation of Granger causality in 7 out of 41 markets (17%) should be interpreted carefully. Granger causality is a conservative statistical test and it requires a strong and consistent temporal signal to reject the null hypothesis of no predictive improvement over the autoregressive baseline. Markets that failed the Granger test are not necessarily markets where Reddit has no relationship with bookings; in many cases the test is underpowered due to insufficient Reddit discussion volume or too short a time series.

The five markets with confirmed clean unidirectional causality represent the highest-confidence deployment targets. For these markets, there is formal statistical evidence that Reddit engagement contains information about future bookings not already captured in booking history. Two of those country-level markets are notable inclusions: neither was among the top correlation markets, yet both passed the Granger test, suggesting that even modest Reddit signals can be statistically significant for smaller markets with sufficient signal consistency.

The bidirectionality finding for two countries deserves attention. The fact that bookings also Granger-cause Reddit engagement for these markets does not invalidate the Reddit signal but it complicates the interpretation. One plausible explanation is that these are destinations where travellers are highly engaged with Reddit communities specifically around the time of booking and travel, creating a feedback loop. Such communities may have travellers share booking confirmations, ask for advice immediately after booking, and post reviews shortly after travel. This creates a simultaneous correlation between bookings and subsequent Reddit activity that the Granger test detects as reverse causality.

Relevance Score

The relevance score's $AUC=0.842$ on high-signal countries under the realistic evaluation framework (top-33% booking days within 28 days) is a strong result in the context of social media-based demand prediction. For comparison, pure chance achieves $AUC=0.5$, and a perfect predictor would achieve $AUC=1.0$. An AUC of 0.842 means that when the relevance score assigns a higher value to country-day A than country-day B, it is correct 84.2% of the time in predicting which will have above-average bookings within 28 days. This is sufficient discriminative power for practical operational use.

The monthly heatmap of relevance scores offers perhaps the most intuitively useful output of the entire analysis. It shows not just whether Reddit predicts bookings, but which destinations are currently elevated, how that elevation has evolved month by month, and which markets show persistent versus episodic relevance. The score of 0.74 in June 2025 corresponds to a period of exceptional growth in that market's tourism

visibility on social media, coinciding with broader media coverage of that market as an emerging European destination. The December 2025 spike may reflect the growing popularity of winter sun destinations among Northern European Reddit communities in the year-end planning period.

Comparison of the Three Scoring Approaches

The three-approach design produced a clear and interpretable result. Approach C (momentum) won under the strict spike prediction framework because momentum signals are well-calibrated for detecting sudden changes. A destination's Reddit buzz jumping sharply above baseline is a relevant signal for predicting isolated booking spike events. However, this came at the cost of being less useful for the broader demand elevation task.

Approach A (weighted linear) won under the realistic framework because sustained engagement signals are better aligned with sustained demand elevation. The evidence-based weights derived from deseasonalized correlations proved to be the appropriate design choice, not because they were theoretically optimal but because they reflected what the data actually showed about which features carry genuine predictive information.

Approach B (sentiment-first) sat between the two in all evaluations, confirming the intermediate importance of sentiment. The finding that emphasising sentiment improves over pure engagement ($B > C$ on the realistic framework), but not over a balanced combination ($A > B$) suggests the optimal scoring architecture is one that uses engagement volume as the primary driver with sentiment as a meaningful secondary modifier; which was the structure of Approach A.

The correlation between Approach A and B scores was 0.925, while both had lower correlation with Approach C (0.736 and 0.708 respectively). This confirms that A and B are measuring similar underlying phenomena, sustained engagement and sentiment level, while C measures something different: momentum and change. All three approaches contribute unique information, and a future iteration of the relevance score could explore whether a four-component approach that explicitly separates level, momentum, and sentiment signals produces further improvement.

VI. ATTEMPT 2 - ENSEMBLE MODEL WITH SHAP ATTRIBUTION

Informed by the work of Attempt 1, this method uses social media data (with a previously confirmed relationship to airline booking data) to predict surges in booking activity as well as to understand the relevance of specific content, themes, and platform data on passenger growth and surges.

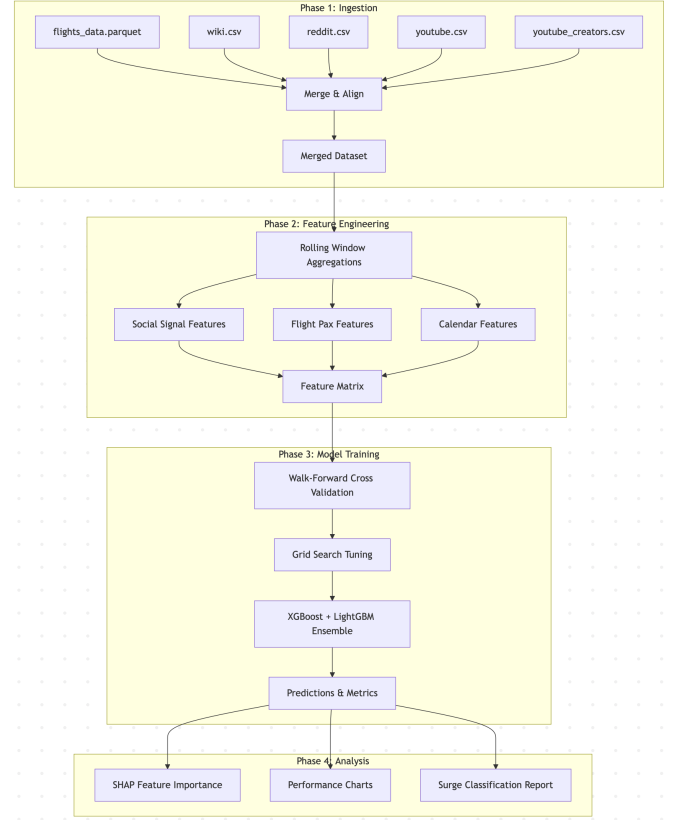


Fig. 16. High level functional diagram outlining the process

A. Data Ingestion and Alignment

We ingest all previously described data, YouTube, Reddit, Wikipedia, Google Trends, and Flight bookings. All data is aligned at a daily level per associated flight booking destination airport. Binary coverage indicators are added per source. Filters are applied to the dataset to exclude destinations with social media coverage and annual pax in an attempt to remove noise from low information airports.

B. Feature Engineering

We engineer numerous features, exploded themes, and booking intent for social data for visibility into features hidden in daily aggregations.

TABLE II. Description of features used for each data source group

Group	N	Feature Types
A. Booking Momentum	10	Rolling mean (7/14/28d) Rolling std (7/14/28d) Momentum ratio (7d/28d) Z-score Above-average flag Destination scale (lagged)
B. Wikipedia	13	Rolling level (3/7/14/28d) Momentum (7d/28d, 3d/7d) Z-score (7d, 28d) Lag ratios (3/7/14/21d)
C. Google Trends	15	Web & YT search level Momentum & fast momentum Z-score (7d, 28d) Lag ratios (3/7/14/21d)
D. Reddit	152	Rolling volume & engagement Sentiment (mean, weighted, velocity) Intent-level sent. (4 intents) Theme counts (10 themes) Theme sentiment (32 themes) Buzz share ratio Consistent sentiment Engagement per post Lag ratios (3/7/14/21d)
E. YouTube	327	Rolling views & virality Shorts vs Videos (separate) Intent-level sentiment (4 intents) Theme counts (10 themes) Theme sentiment (32 themes) Video tag groups (4 groups: booking intent, destination discovery, events, accommodation) Buzz share ratio Top-video features Intersection features Commercial intent Consistent sentiment
F. Cross-Source	7	Social acceleration count Leads-pax flags (3 sources) Social leading score Sources elevated count Mean source z-score
G. Calendar	6	Month, day of week Week of year, is weekend Cyclical encoding (sin/cos)
Target	1	Growth Rate (Pax) $\frac{\text{forward_7d_mean_deseas} - \text{baseline_28d_mean_deseas}}{ \text{baseline_28d_mean_deseas} }$ clipped to $[-2, 5]$ to reduce noise Surge = growth_rate > 0.75

We use a series of short, medium and long preceding rolling means and lag ratios to capture the impact of previous social media activity on bookings. Zero variance features are screened and removed from the dataset.

The approach to deseasonalisation leans on removing day of week effects primarily, as monthly deseasonalisation is unreliable with only 12 months of data. Deseasonalisation is performed using a rolling expanding mean per weekday per destination. We capture seasonal information by capturing calendar features in the model.

A log transformation is applied to a large number of social media metrics to reduce skew and allow higher information extraction from these features by the model

C. Model Architecture

The model performs regression to predict the pax growth rate, before thresholding is applied and a particular day is classified as a surge. An ensemble of XGBoost + LightGBM predicts the continuous growth rate, this method was chosen

over a classifier to preserve surge magnitude information, avoiding hard binary decisions on the data. Using the 75% growth rate threshold for surge classification finds 20% of the data set classified as a surge period.

Two configurations are trained, a full model and social model. The social model excludes pax momentum features, in an attempt to answer whether social media alone is able to independently predict surges in bookings.

The ensemble of XGBoost and LightGBM models provides stability improvements over a single model, LightGBM effectively handles sparse datasets like the one used in the problem.

D. Walk-Forward Cross-Validation

Walk forward cross validation uses a 90-day initial training window, expanding by 14-day steps, ensuring the model does not look forward when training. Out of fold predictions are then assembled across the entire timeline for evaluation.

E. Hyperparameter Tuning

The model hyperparameter space is explored using a grid search on 2 key parameters: learning rate, max depth. 6 combinations across 5 different ensemble weights are evaluated across 5 walk forward folds. The number of estimators is capped using the early stopping parameter, terminating the training process when performance stops improving.

F. Feature Importance and SHAP

SHapley Additive exPlanations are used to understand the model outputs and feature importances. This provides both local and global interpretability to the model, allowing us to understand the model's behavior as well as to understand what impacted each feature on specific model predictions.

SHAP values are grouped by source and summed to quantify each source's overall aggregate contribution. Per surge attribution is performed for each true-positive surge, this provides explainability by the model and allows users to understand which content and sources may have contributed to a surge.

G. Results and Evaluation

1) *Model Performance:* The model's success in predicting and identifying surges is moderate but reasonable considering the human nature of predicting flight bookings, the full model (including pax and pax momentum features) performs better than the social only model. This is to be expected considering the continuous nature of airline bookings and particularly their typical 'ramp up' behavior. Hyperparameter tuning was performed via grid search over learning rate, tree depth, and number of estimators (18 combinations across 5 folds), with ensemble weights optimised per model. The **calendar model** includes the 6 calendar features and destination scale. The **social model** (Reddit, YouTube, Wikipedia, Google Trends, and Calendar) includes social features and destination scale (a long lagged average passengers to indicate scale at the destination). The social model benefits the most from grid search hyperparameter tuning, likely due to its noisy features,

seeing a 23% relative gain to R^2 to the default parameters. The **full model** includes all calendar, social and pax booking momentum features.

From the results, we see that the surge prediction of the model mostly lags actual surges. The model aims to predict

TABLE III
REGRESSION PERFORMANCE — GROWTH RATE PREDICTION (TUNED)

Model	RMSE ↓	MAE ↓	R^2 ↑
Calendar	1.4436	1.0568	0.0432
Social	1.3889	0.9843	0.1144
Full	1.2756	0.8580	0.2529
<i>Incremental gains</i>			
Social — Calendar	−0.0547	−0.0725	+0.0712
Full — Social	−0.1133	−0.1263	+0.1385
Full — Calendar	−0.1680	−0.1988	+0.2097

the growth rate in passengers at a particular airport for each day, which is given by the formula:

$$\frac{\text{forward_7d_mean_deseas} - \text{baseline_28d_mean_deseas}}{|\text{baseline_28d_mean_deseas}|}$$

The result of the regression analysis shows modest predictive power considering its attempt to predict a noisy, continuous target influenced by a wide range of external factors. The classification aims to predict ‘surges’ in bookings, specified as growth rates greater than 75%, which represents approximately 20% of booking data. Although modest, the models provide meaningful regression results that improve consistently as more signal layers are added. We can see across the models that as more features are added over the calendar baseline R^2 score improves. The R^2 score of the model improves progressively from calendar → social → full, with social features nearly tripling the explained variance from 0.043 to 0.114. This implies that social signals meaningfully improve the model’s ability to predict the magnitude and timing of booking surges. The inclusion of pax features to create the full model explains roughly a quarter of the variance in growth rate ($R^2 = 0.253$), and social features account for 34% of the total R^2 gain from calendar to full (+0.071 of +0.210), demonstrating a substantial contribution from contextual data. We’re able to better understand performance once we classify

TABLE IV
SURGE CLASSIFICATION PERFORMANCE (THRESHOLD: 75TH PERCENTILE, SURGE RATE $\approx$ 20.1%). PR-AUC IS THE PRIMARY METRIC GIVEN CLASS IMBALANCE (RANDOM BASELINE $\approx$ 0.20).

Model	PR-AUC	ROC-AUC	F1	Precision	Recall
Calendar	0.3092	0.6659	0.3821	0.2734	0.6345
Social	0.3710	0.7065	0.3989	0.3352	0.4924
Full	0.4936	0.7535	0.4879	0.5207	0.4590
<i>Incremental gains</i>					
Soc. — Cal.	+0.0618	+0.0406	+0.0168	+0.0618	−0.1421
Full — Soc.	+0.1226	+0.0470	+0.0890	+0.1855	−0.0334
Full — Cal.	+0.1844	+0.0876	+0.1058	+0.2473	−0.1755

surges from the regression’s predicted growth rate. With tuned

hyperparameters, the social model achieves a PR-AUC of 0.371, a 20% improvement over the calendar baseline (0.309). This is a notable gain compared to the untuned run, where the social increment was only 10.7%, suggesting that tuning disproportionately benefited the social model by allowing it to better exploit the noisier, higher-dimensional social feature space. PR-AUC results are focused on due to the skewed nature of the data, with only 20% being classified as surges. Social features now account for 36.5% of the total PR-AUC gain from calendar to full (+0.066 of +0.180), up from 21.6% in the untuned configuration. Again, we see that the social model improves on the calendar model, and that the inclusion of pax momentum features specifically sharpen predictions near surge thresholds — the full model’s precision (0.521) is nearly double the calendar’s (0.273), while recall converges across models.

2) *SHAP Source Contribution*: The SHAP source contribution forms the basis for the **relevance score**, used to understand the contributions from each social media source to a predicted surge in bookings.

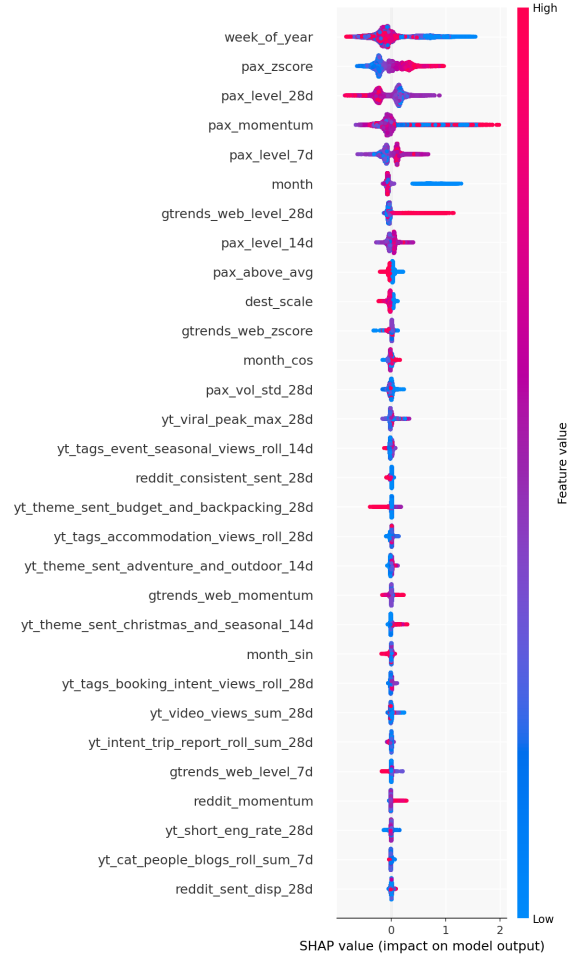


Fig. 17. SHAP feature importance for the full model.

We observe that the top 6 features in the full model are pax and calendar related, with pax momentum having the greatest

SHAP spread. Intuitively, this makes sense considering the 'ramp' up behaviour of booking surges, the model learns this. The asymmetry of the month value is also expected as travel increases for European summer. There are a number of high ranking social features, but they have a lower impact on the model than the dominating passenger features.

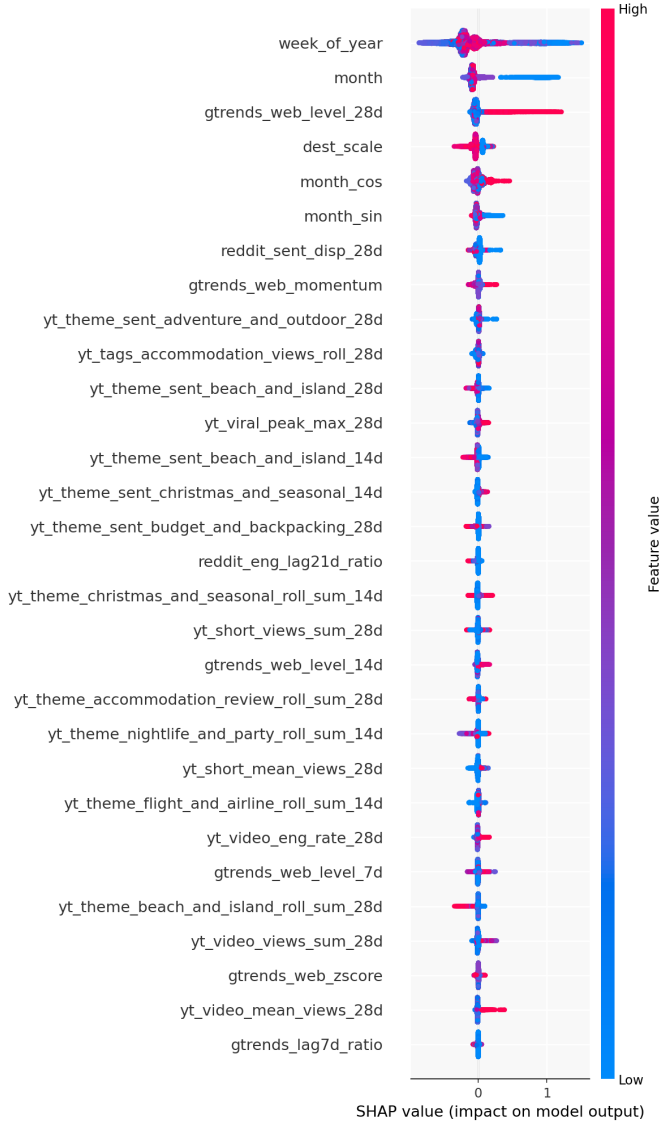


Fig. 18. SHAP feature importance for the social model.

Here we see the nuance in contribution of the social media features, YouTube theme sentiment features are more prominent, validating the NLP and theme extraction as a valuable exercise. Reddit sentiment dispersion ranks 7th, indicating that disagreement and therefore discourse is informative.

a) *City Y Destination Case Study:* The full model tracks Country Y's surge patterns reasonably well, capturing the May peak as well as the Christmas peaks later in the year.

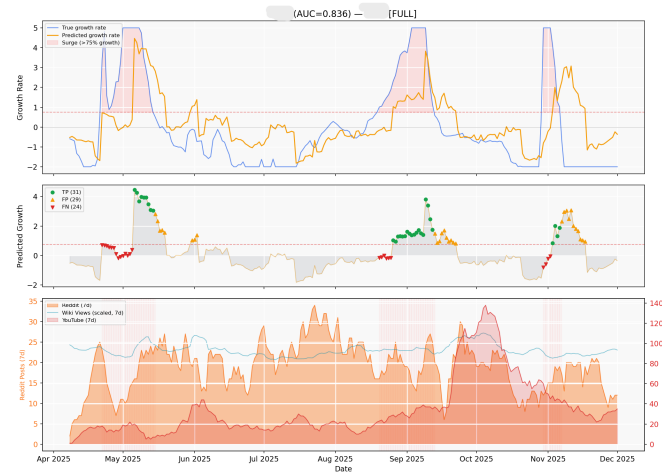


Fig. 19. City Y booking surge predictions using the full model, tracking three distinct surge periods.

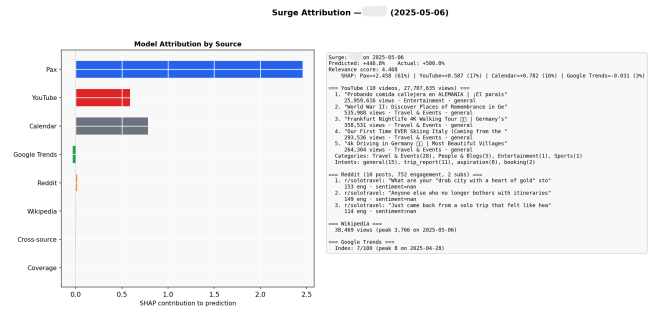


Fig. 20. City Y booking surge feature and content attribution,

b) *City Z Destination Case Study:* The model predicts many false positives for City Z as the passenger growth hovers near the target 75%, missing a number of actual surges. The model captures the surge later in the year and we are able to interrogate the social media that contributed to this.

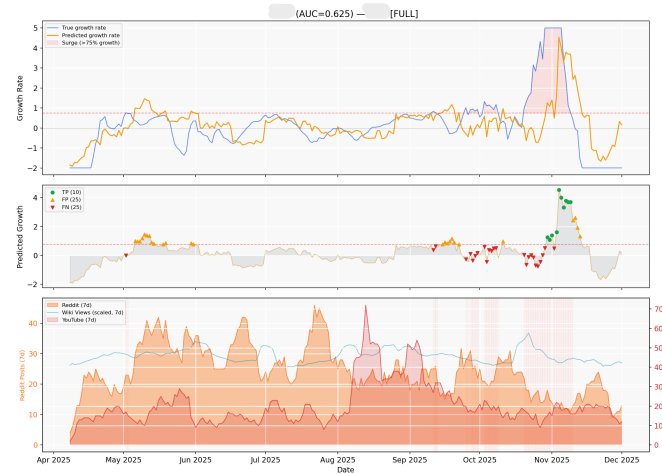


Fig. 21. City Z booking surge predictions using the full model, presenting a more difficult prediction case than City Y.

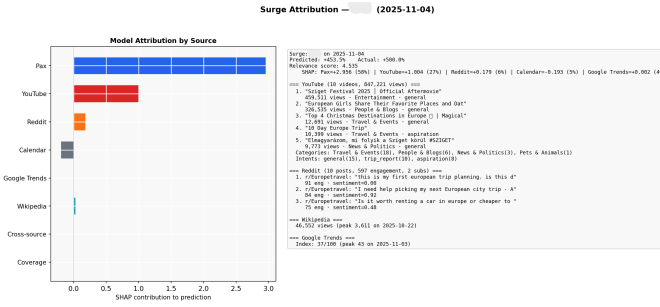


Fig. 22. City Z booking surge feature and content attribution,

We see that both Reddit and YouTube can be used to primarily explain the surge. Looking into the related videos, we see contributions relating to the Sziget music festival and the theme of Christmas.

3) Content Understanding:

a) *Topical Triggers*: After using the model to identify surges, we're able to perform postprocessing on the original YouTube and Reddit datasets to identify recent, potentially influential themes and information. Thanks to the theme information earlier extracted using NLP for both Reddit and YouTube data, we're able to correlate true-positive surges to understand the most impactful themes on surges in booking activity across destinations.

We list out popular 'themes' from our list of 32:

TABLE V
TOP TRAVEL THEMES ACROSS ALL SURGE ATTRIBUTIONS

Travel Theme	Surges	Videos	Views	Sent.
Packing & Planning	1,489	12,875	1,408M	0.62
History & Culture	1,356	6,159	207M	0.65
Adventure & Outdoor	1,330	6,316	476M	0.65
Food & Drink	1,322	7,608	482M	0.72
General Travel	1,241	4,962	567M	0.37
Beach & Island	1,039	6,778	363M	0.66
Festival & Events	989	6,785	655M	0.52
Flight & Airline	922	4,957	157M	0.53
Christmas & Seasonal	896	5,041	215M	0.68
Accommodation Review	894	1,999	84M	0.66
City Break	845	2,168	64M	0.59
Family Travel	797	1,746	50M	0.77
Photography & Scenic	621	1,095	327M	0.76
Train & Rail Travel	511	893	10M	0.44
Road Trip & Driving	501	795	39M	0.52
...	...	...	...	...
Wellness & Spa	123	195	0.4M	0.39
Sports & Fitness	13	13	0.1M	-0.17
Wildlife & Nature	2	2	<0.1M	-0.08

We're able to see that many expected travel related themes can be attributed to surge events. The greatest contributor, in terms of correlated surges and views is packing and planning content. This makes intuitive sense, as this content can form the basis for planning trips. We see low attribution to surges and low sentiment for wellness and spa, sports and fitness and wildlife and nature content.

b) *Influencer Identification*: Across the datasets, the concept of an influencer only applies to the YouTube dataset. Using the same method as applied to Topical Triggers, we

can correlate true-positive surge predictions to understand the most influential channels.

TABLE VI
CHANNEL STATISTICS

Channel	Surges	Dests	Views	Subs
DW Travel	101	10	37,742,622	409,000
BonjourCity	94	8	782,446	33,900
Jan and Ann	92	10	19,284,015	226,000
Luke Patrick Travel	91	10	895,953,944	131,000
Daidsbeenhere	88	4	13,782,623	1,410,000
4K MAGIC WORLD	85	5	25,662,906	5,840
CIAO ITALY	84	7	18,192,663	135,000
Safari	83	1	12,445,717	3,150,000
DoTravel	79	11	983,785	77,200
TravelScout	79	13	415,694	42,700

VII. LIMITATIONS

This project faced several practical challenges related to data collection. Accessing social media APIs was either not feasible or slow, which restricted the data volume and variety. We were able to work around these limitations by alternative scraping methods and working within minimum API tier limits, which we were approved for, such as for Wikimedia.

Another significant limitation in this project was the limited subreddit selection. The Reddit data used in this project was sourced exclusively from five English-language travel communities: r/EuropeTravel, r/UKTravel, r/SoloTravel, r/Travel, and r/Backpacking. While these subreddits represent some of the largest and most active travel communities on Reddit, they capture only a fraction of the online spaces where meaningful travel inspiration and destination discussion takes place. Within Reddit itself, a large volume of travel-relevant discussion occurs in destination-specific subreddits, where users ask specific questions about cities, neighbourhoods, accommodation, and activities. These communities were not included in the analysis, meaning destination-specific viral moments that originated in niche subreddits rather than general travel communities would be invisible to the model.

Reddit's user base introduces a further representational limitation. Reddit generally overrepresents younger, English-speaking, technically literate users. This is a demographic profile that may not uniformly represent LHG's full customer base, which includes older travellers, non-English speakers, families, and business travellers who may rely on entirely different information channels when making travel decisions. The signal captured here is therefore a partial view of social media-driven travel inspiration, most relevant for the subset of LHG's customers who use English-language Reddit communities as part of their travel planning process.

Additionally, the LHG booking dataset covers only 12 months of transaction data, spanning December 2024 to December 2025. This single-year window introduced several constraints that limited both the depth of the analysis and the robustness of its conclusions. Most significantly, 12 months of data represents only a single seasonal cycle. The STL decomposition applied throughout this analysis used a period

of 7 days to remove the weekly booking cycle which was the most granular and consistently observable seasonal pattern within a one-year window. While removing weekly seasonality produced substantially improved correlation results, the analysis was not able to address annual seasonality with the same rigour.

The short data window also constrained the statistical power of the Granger causality tests. Each country's time series contained only a year worth of observations after joining, which is a relatively short series for a multivariate time series test that consumes degrees of freedom for each lag tested. This means the Granger test was likely underpowered for markets with lower signal-to-noise ratios, and that some of the 34 markets showing no significant causality may in fact have genuine Reddit-to-bookings relationships that a longer series would have been able to detect. With 2-3 years of booking data the number of markets showing confirmed Granger causality would be expected to increase.

VIII. CONCLUSION

A. Attempt 1 - Reddit Signal & Flight Booking Interactions and Relevance Scoring

The central finding is that Reddit engagement is a genuine but modest and market-specific predictor of booking demand. After removing the shared weekly seasonal patterns that inflated raw correlations by approximately 60%, the genuine deseasonalized predictive signal (a Spearman correlation of $r=0.201$ between 7-day rolling Reddit engagement and deseasonalized bookings at a 7-day lag) was modest in absolute terms but statistically meaningful and temporally coherent. The decay of this correlation with increasing lag, from $r=0.201$ at lag 7 to $r=0.116$ at lag 28, confirmed that the relationship is genuinely temporal rather than spurious, establishing an actionable commercial window of approximately 1-2 weeks.

Secondly, sustained engagement consistently outperformed both sentiment level and engagement momentum as a predictor. A destination that has accumulated 7-14 days of above-baseline Reddit discussion is a stronger booking signal than one that experienced a single viral post. Monitoring rolling engagement trends rather than reacting to daily spikes is the operationally sound approach.

Lastly, consistent sentiment outperformed raw sentiment by a substantial margin after deseasonalization, with consistent sentiment retaining $r=0.147$ across all tested lag windows while mean daily sentiment collapsed to $r=0.018$. This confirms that the sustained positive reputation of a destination within travel communities is what matters, not any individual day's emotional tone.

The pipeline also demonstrated the value of comparing multiple analytical approaches rather than committing to a single method. The three relevance score approaches (weighted linear, sentiment-first, and engagement momentum) each illuminated a different aspect of the signal, and their comparative evaluation revealed that the evidence-based weighted linear approach was the most appropriate for predicting sustained demand elevation while the momentum approach was better

calibrated for rare spike detection. This comparison, rather than any single result, provides the most complete picture of how the Reddit signal can be operationally deployed across different commercial use cases.

B. Attempt 2 - Ensemble Model with SHAP Feature Attribution

The regression performance ladder improvements shows the predictive power of social signals (Reddit, YouTube, Wikipedia and Google Trends information) when aiming to predict booking surges. Although predictions are, the non-negligible and worth further exploring.

From this model, we have been able to utilise SHAP interpretability to understand the contributing features and themes behind predicted surges. We note that 'travel' related content is a popular contributor to surges, as well as popular travel channels such as the German DW Travel.

The model is able to attribute 'relevance' to each model feature using the SHAP attributions, also determining the percentage impact of each data source to true positive surges.

C. Final Remarks

Overall in this project, the collected data provides evidence that social media has some meaningful signal about future airline booking demand. After both attempts, the results suggest that higher online attention provides predictive value to predicting booking volume. The analysis of Reddit data explored how sustained engagement and positive engagement is a better signal than a few viral posts. The ensemble model approach that combined all of the gathered data and booking contexts also showed how that data can improve predictive performance of booking volume. Although the model is not perfect at predicting surges, this evidence justifies further exploration and gathering more data to improve results. With additional data and refinement, this work sets a strong foundation to use social and online media as a signal for predicting booking surges.

IX. RECOMMENDATIONS

Although correlations and model results are somewhat weak, they show a definite relationship between social media data and airline destination booking data. This result is highly interesting and worth exploring further.

Potential future improvements to predicting and attributing surges:

- At least 2 years of booking data, allowing full monthly STL decomposition
- Expand the Reddit and YouTube datasets to increase coverage and statistical power
- Explore other model types for predicting surges, including neural nets

X. WORKLOAD DISTRIBUTION

All group members contributed in meaningful, and creative ways to the development of the project. The team enjoyed the process of collaboration, from the collection of data, to processing through to evaluation and analysis.

TABLE VII
WORKLOAD DISTRIBUTION

Name	Contribution
Mohammad Yousuf Qureshi	33%
Rikesh Patel	33%
Wade Geles	33%

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